White Matter Hyperintensity volumes (global and regional) using the cross-sectional Bayesian Model Selection (BaMoS) pipeline - Insight46 phase 1

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Contents

Document details

Categories of variables	Imaging outcome measure – white matter hyperintensity volumes (global and regional) generated using the cross-sectional Bayesian Model Selection (BaMoS) pipeline
Summary of work undertaken	Values were generated using the BaMoS pipeline. The WMH segmentations were then QC'd by CL – further detail is provided below.
Names of source variables	Variables, are, specified, below
Output variables	bamosuseable_i46p1, lesiontot_i46p1, distwmhocc_i46p1, distwmhfront_i46p1, distwmhpar_i46p1, distwmhbg_i46p1, distwmhtemp_i46p1, distwmhlayer1_i46p1, distwmhlayer2_i46p1, distwmhlayer3_i46p1, distwmhlayer4_i46p1, volwmhocc_i46p1, volwmhfront_i46p1, volwmhpar_i46p1, volwmhtemp_i46p1, volwmhbg_i46p1, volwmhlayer1_i46p1, volwmhlayer2_i46p1, volwmhlayer3_i46p1, volwmhlayer4_i46p1, percentwmhocc_i46p1, percentwmhfront_i46p1, percentwmhpar_i46p1, percentwmhtemp_i46p1, percentwmhbg_i46p1, bamosfailreason_i46p1
Papers using these variables	Lane et al 2019 Investigating the associations between blood pressure across adulthood and late-life brain structure and pathology in the 1946 British birth cohort: an epidemiological study. Lancet Neurology

BaMoS outcome variables at 69-71y (Phase 1 Insight 46) (2019-CL)

This document details all the variables created from the BaMoS pipeline and how they were cleaned (i.e. BaMoS QC).

BaMoS QC

Cross-sectional BaMoS processing1 was performed using all available T1 and FLAIR images and the output variables extracted. BaMoS segmentations were then visually reviewed in NiftyMidas, overlaid on the resampled FLAIR image.

Issues with segmentation that arose e.g mis-segmentation of cortical strokes, excessive temporal lobe artefacts, were flagged and the scans re-run with modified BaMoS scripts, the segmentations for which were then returned and re-reviewed. Choroid plexus which was excessively mis-segmented (as determined on visual review) was manually edited in NiftyMidas and updated outputs generated.

Certain pathologies e.g. demyelination, some cortical strokes, vascular abnormalities causing white matter hyperintensities (WMH) not related to cerebral small vessel disease, that were inappropriately segmented were excluded as necessary. Individuals with neurological diagnoses were not otherwise excluded.

bamosfailreason_i46p1 – the reason why a segmentation was not useable (only relevant if bamosuseable_i46p1 = 0)

Code	Value
1	T1 QC fail
2	FLAIR QC fail
3	BaMoS segmentation fail
4	Cortical stroke inappropriately segmented as WMH
5	Demyelination inappropriately segmented as WMH
6	Other vascular pathology inappropriately segmented
-99	Not applicable (either bamosuseable=1 or no scan available)

BaMoS outcome variables

Generated outcomes from BaMoS segmentations. For use in analyses, volumetric measures should be adjusted for head size using the total intracranial volume (TIV) measure (**variable x**, **detailed in document x**). It is recommended that this is done by adjusting for TIV within a statistical model, rather than creating a new proportional value "the proportion method".2,3 Regional lobar values are derived using the GIF v.3 parcellations of the volumetric T1 image4. Regional layer values are derived by dividing the subcortical region into 4 equidistant layers according to the normalised distance between the ventricular system and the white matter/cortical grey matter interface adapted from the method described by Yezzi et al to compute cortical thickness5.

Summary statistics

bamosuseable_i46p1	Freq.	Percent
No scan available (-99)	31	6.18
Not useable (0)	16	3.19
Useable (1)	455	90.64

bamosfailreason_i46p1	Frequency	Percent
Not applicable (-99)	486	96.81
T1 fail (1)	<10	<2
FLAIR fail (2)	<10	<2
Bamos segmentation fail (3)	<10	<2
Cortical stroke mis-segmented (4)	<10	<2
Demyelination mis-segmented (5)	<10	<2
Other vascular anomaly mis-segmented (6)	<10	<2

Variable	Obs	Mean	Std. Dev.	Min	Max
lesiontot_I46P1	455	5.109443	5.441696	.2661919	33.66793
volwmhocc_i46p1	455	.654722	.6169186	0	5.820504
volwmhfront_i46p1	455	2.55895	2.91357	.0847699	18.22171
volwmhpar_i46p1	455	1.069738	1.578511	.0013348	10.0226
volwmhtemp_i46p1	455	.53425	.6169533	.0163922	4.18243
volwmhbg_i46p1	455	.291783	.3828594	.0013348	2.595435
volwmhlayer1_i46p1	455	1.680457	1.406311	.0702311	8.167364
volwmhlayer2_i46p1	455	1.317616	1.739918	.0323781	14.00983
volwmhlayer3_i46p1	455	1.025574	1.537759	.0136052	10.76657
volwmhlayer4_i46p1	455	1.085797	1.329666	.0319939	9.112107
distwmhocc_i46p1	455	.1749019	.1068451	0	.5984535
distwmhfront_i46p1	455	.4905785	.1389233	.1478342	.8764374
distwmhpar_i46p1	455	.1587049	.0957124	.0014029	.5342677
distwmhbg_i46p1	455	.0647237	.0435142	.0021898	.3443137
distwmhtemp_i46p1	455	.1110911	.0573211	.0132275	.3828413
distwmhlayer1_i46p1	455	.394021	.1336008	.0945299	.7496052
distwmhlayer2_i46p1	455	.2259022	.0772718	.0515649	.4997641

distwmhlayer3_i46p1	455	.1653936	.0707513	.0219116	.4312162
distwmhlayer4_i46p1	455	.2146832	.1054891	.0446742	.647063
percentwmhocc_i46p1	455	1.135649	.9654228	0	7.283258
percentwmhfront_i46p1	455	1.249054	1.406807	.0381781	9.244668
percentwmhpar_i46p1	455	1.113462	1.607811	.0016583	9.820433
percentwmhtemp_i46p1	455	.5646139	.6478538	.0179838	4.883367
percentwmhbg_i46p1	455	.7077708	.8949738	.0032964	6.353084

References

1 Sudre CH, Cardoso MJ, Bouvy WH, Biessels GJ, Barnes J, Ourselin S. Bayesian model selection for pathological neuroimaging data applied to white matter lesion segmentation. IEEE Trans Med Imaging [online serial]. 2015;34:2079–2102. Accessed at: http://ieeexplore.ieee.org/document/7078891/. Accessed June 25, 2017.

2 Malone IB, Leung KK, Clegg S, et al. Accurate automatic estimation of total intracranial volume: a nuisance variable with less nuisance. Neuroimage. Elsevier; 2015;104:366–372.

3 Barnes J, Ridgway GR, Bartlett J, et al. Head size, age and gender adjustment in MRI studies: a necessary nuisance? Neuroimage [online serial]. 2010;53:1244–1255. Accessed at: http://www.ncbi.nlm.nih.gov/pubmed/20600995. Accessed September 1, 2017.

4 Cardoso MJ, Modat M, Wolz R, et al. Geodesic Information Flows: Spatially-Variant Graphs and Their Application to Segmentation and Fusion. IEEE Trans Med Imaging. 2015;34:1976–1988.

5 Yezzi AJ, Prince JL. An eulerian pde approach for computing tissue thickness. IEEE Trans Med Imaging [online serial]. 2003;22:1332–1339. Accessed at: http://www.ncbi.nlm.nih.gov/pubmed/14552586. Accessed August 13, 2018.

Variables in this document

26 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › White matter hyperintsity (WMH) volumes [60332] › White matter hyperintsity (WMH) volumes - Cross-sectional [603321]		
bamosfailreason_i46p1	Reason why a segmentation was not useable (only relevant if bamosuseable_i46p1=0) - Cross-sectional - Insight46 phase 1	topsheet field
bamosuseable_i46p1	Whether the white matter hyperintensity (WMH) segmentation is useable - Cross-sectional - Insight46 phase 1	topsheet field
distwmhbg_i46p1	Proportion of global white matter hyperintensity (WMH) in the basal ganglia (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field
distwmhfront_i46p1	Proportion of global white matter hyperintensity (WMH) in the frontal lobes (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field
distwmhlayer1_i46p1	Proportion of global white matter hyperintensity (WMH) in layer 1 (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field
distwmhlayer2_i46p1	Proportion of global white matter hyperintensity (WMH) in layer 2 (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field
distwmhlayer3_i46p1	Proportion of global white matter hyperintensity (WMH) in layer 3 (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field
distwmhlayer4_i46p1	Proportion of global white matter hyperintensity (WMH) in layer 4 (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field

Variable	Label	How it was found
distwmhocc_i46p1	Proportion of global white matter hyperintensity (WMH) in the occipital lobes (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field
distwmhpar_i46p1	Proportion of global white matter hyperintensity (WMH) in the parietal lobes (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field
distwmhtemp_i46p1	Proportion of global white matter hyperintensity (WMH) in the temporal lobes (value 0 – 1) - Cross-sectional - Insight46 phase 1	topsheet field
lesiontot_i46p1	Global white matter hyperintensity (WMH) volume (ml). Includes subcortical grey matter but not the infratentorial region - Cross-sectional - Insight46 phase 1	topsheet field
percentwmhbg_i46p1	Percentage of total volume in basal ganglia considered white matter hyperintensity (range 0 – 100) - Cross-sectional - Insight46 phase 1	topsheet field
percentwmhfront_i46p1	Percentage of total white matter in the frontal lobes considered to be white matter hyperintensity (range 0 – 100) - Cross-sectional - Insight46 phase 1	topsheet field
percentwmhocc_i46p1	Percentage of total white matter in the occipital lobes considered to be white matter hyperintensity (range 0 – 100) - Cross-sectional - Insight46 phase 1	topsheet field
percentwmhpar_i46p1	Percentage of total white matter in the parietal lobes considered white matter hyperintensity (range 0 – 100) - Cross-sectional - Insight46 phase 1	topsheet field
percentwmhtemp_i46p1	Percentage of total white matter in the temporal lobes considered white matter hyperintensity (range 0 – 100) - Cross-sectional - Insight46 phase 1	topsheet field
volwmhbg_i46p1	White matter hyperintensity (WMH) volume (ml) in the basal ganglia - Cross-sectional - Insight46 phase 1	topsheet field
volwmhfront_i46p1	White matter hyperintensity (WMH) volume (ml) in the frontal lobes - Cross-sectional - Insight46 phase 1	topsheet field
volwmhlayer1_i46p1	White matter hyperintensity (WMH) volume (ml) in layer 1 (inner) - Cross-sectional - Insight46 phase 1	topsheet field
volwmhlayer2_i46p1	White matter hyperintensity (WMH) volume (ml) in layer 2 - Cross-sectional - Insight46 phase 1	topsheet field
volwmhlayer3_i46p1	White matter hyperintensity (WMH) volume (ml) in layer 3 - Cross-sectional - Insight46 phase 1	topsheet field
volwmhlayer4_i46p1	White matter hyperintensity (WMH) volume (ml) in layer 4 (outer) - Cross-sectional - Insight46 phase 1	topsheet field
volwmhocc_i46p1	White matter hyperintensity (WMH) volume (ml) in the occipital lobes - Cross-sectional - Insight46 phase 1	topsheet field
volwmhpar_i46p1	White matter hyperintensity (WMH) volume (ml) in the parietal lobes - Cross-sectional - Insight46 phase 1	topsheet field
volwmhtemp_i46p1	White matter hyperintensity (WMH) volume (ml) in the temporal lobes - Cross-sectional - Insight46 phase 1	topsheet field

2 medium-confidence matches

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Heart and circulatory conditions [51014]		
stroke	Have you had a stroke in the last 10 years - at age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan